Euler-Based Simulated Likelihood for Multidimensional Diffusions:

Dimensional Variance, Joint Asymptotics, and an Application Revisited

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Abstract

A widely used simulation method for likelihood inference in discretely observed diffusions approximates each transition density by simulating an Euler scheme to one step before the observed endpoint and averaging the final Gaussian density. This endpoint estimator is a shrinking-bandwidth kernel estimator whose Monte Carlo variance grows with the discretisation level at a dimension-dependent rate: in K dimensions the second moment of one simulated contribution is of order $M^{K/2}$, so the variance is of order $M^{K/2}/S$ rather than uniformly $1/S$. We derive exact Brownian moments, the corrected central-limit normalisation $\sqrt{S/M^{K/2}}$, the resulting bias–variance trade-off with optimal rate $S^{-4/(K+4)}$, and the scale of the nonlinear log-likelihood error. We then construct a joint-limit counterexample: along $M = S \to \infty$, which satisfies the rate condition $\sqrt{S}/M \to 0$ of Brandt and Santa-Clara (2002) and their own regularity assumptions, the simulated density converges in probability to zero for every $K > 2$ although the true density is strictly positive. In four dimensions their condition is equivalent to $S/M^2 \to 0$, whereas mean-square consistency requires $S/M^2 \to \infty$. We then examine the mathematical status of the four-dimensional exchange-rate application: square-root boundaries, the reality of the incompleteness state along simulated paths, an implicitly defined volatility process whose reported loadings admit either zero or two positive branches, positive-semidefiniteness of the Brownian correlation matrix, a minimum-incompleteness selection rule with a corner solution, and binding inequality constraints. The paper does not assert that the published numerical estimates are incorrect; it establishes that their stated asymptotic justification is incomplete and, in the four-dimensional case, contradicted by an exact counterexample.

Keywords: simulated maximum likelihood; diffusion processes; Euler scheme; transition density; Monte Carlo variance; curse of dimensionality; diffusion bridges; nonregular inference.

JEL classifications: C13; C15; C32; G12; G15.

1 Introduction

Continuous-time diffusion models are often observed only at discrete dates. Their exact transition densities are rarely available, which makes exact maximum likelihood difficult even when simulation of approximate sample paths is straightforward. One influential response is the simulation method introduced by Pedersen (1995a) and Pedersen (1995b) and developed and applied by Brandt and Santa-Clara (2002). The method divides each observation interval into M Euler substeps, simulates the process through the first $M - 1$ steps, evaluates the

Gaussian density of the final Euler step at the observed endpoint, and averages this density over S simulated paths.

For fixed M, the construction is elementary: the sample average converges to the transition density of the M-step Euler chain as $S \to \infty$. For fixed S it is also clear that making the final time step smaller turns the Gaussian endpoint density into an increasingly narrow and increasingly high function. The mathematical question is therefore not whether the two separate approximations are individually meaningful. It is whether they remain compatible when M and S increase jointly.

The central observation of this paper is that the final Gaussian density is a kernel with bandwidth $h^{1/2}$, where $h = 1/M$. In K dimensions its effective volume is of order $h^{K/2}$ and its height is of order $h^{-K/2}$. Only simulations that fall in a shrinking neighbourhood of the endpoint contribute materially. The effective number of such simulations is therefore of order

$$Sh^{K/2} = \frac{S}{M^{K/2}}.$$

This quantity, rather than S alone, controls the Monte Carlo approximation.

That the endpoint construction is statistically wasteful is not a new observation. Durham and Gallant (2002) note that unconditioned Euler paths rarely arrive near the observed endpoint and propose bridge-based proposals for that reason; Elerian et al. (2001) augment the path with a Metropolis–Hastings block for the same reason; Stramer and Yan (2007) document numerically that the accuracy of the endpoint estimator deteriorates as M increases with S held fixed; and Lindström (2012) regularises the Pedersen proposal towards a bridge on the same diagnosis. What that literature records is a qualitative and largely numerical finding about efficiency. What is established here is different in kind: an exact dimensional law for the moments of the endpoint summand, the identification of $S/M^{K/2}$ as the quantity that governs the estimator, the corrected triangular-array normalisation, and a joint-limit counterexample showing that the published convergence statement is not merely unproved but false. The distinction matters because an efficiency loss can be absorbed by more simulations, whereas an inconsistency along an admissible rate path cannot.

The paper makes six main contributions.

First, we derive exact finite-M moments for the estimator under K-dimensional Brownian motion. The Euler approximation is exact in this benchmark, so any failure is caused by the endpoint Monte Carlo construction and not by discretisation bias, boundary behaviour, nonlinear coefficients, or parameter identification.

Second, we construct a direct counterexample to the joint convergence result stated in Lemma 2 and Theorem 1 of Brandt and Santa-Clara (2002). For every $K > 2$, choosing $M = S = n$ makes the simulated density at the Brownian origin converge in probability to zero. Standard Brownian motion satisfies the regularity assumptions of that paper exactly, and the sequence satisfies its condition $\sqrt{S}/M \to 0$, so the counterexample cannot be excluded by strengthening the hypotheses. In the four-dimensional application, their condition requires $S/M^2 \to 0$, while the variance calculation requires $S/M^2 \to \infty$ for mean-square consistency.

Third, we establish the generic local moment law for a smooth uniformly elliptic diffusion. If G_h denotes one endpoint-density summand, then for $r > 1$,

$$h^{(r-1)K/2}\mathbb{E}[G_h^r] \longrightarrow (2\pi)^{-(r-1)K/2} r^{-K/2} p(y \mid x)|V(y)|^{-(r-1)/2}.$$

In particular, $\mathrm{Var}(G_h)$ is of order $h^{-K/2}$. This gives the corrected triangular-array central-limit normalisation

$$\sqrt{Sh^{K/2}}\,(\widehat{q}_{M,S} - q_M),$$

not $\sqrt{S}(\widehat{q}_{M,S} - q_M)$.

Fourth, we combine the $O(M^{-1})$ Euler density bias used by Brandt and Santa-Clara (2002) with the corrected simulation variance. The resulting pointwise mean squared error is

$$O(M^{-2}) + O\left(\frac{M^{K/2}}{S}\right),$$

which is minimised at $M \asymp S^{2/(K+4)}$ and yields the rate $S^{-4/(K+4)}$. The estimator therefore exhibits an explicit curse of dimensionality.

Fifth, we trace the density-level result into the log likelihood. The nonlinearity of the logarithm operates on the effective Monte Carlo size $S/M^{K/2}$. Whenever a standard second-order delta expansion is valid, the leading log-density bias is of order $M^{K/2}/S$, rather than uniformly of order $1/S$. This distinction matters because simulated likelihood sums a log transformation over all observations.

Sixth, we examine the mathematical status of the exchange-rate application. We first record that the implemented design does not satisfy the paper's own rate conditions, which jointly require $N^4 \ll S \ll M^2$, by several orders of magnitude. The empirical model then contains square-root diffusions that violate the smoothness and nondegeneracy assumptions of the generic theorem, though we show that for the interest rates this violation is numerically inconsequential at the reported estimates. The market-incompleteness state is not: its Feller ratio equals one half for every parameter value, and because the estimator eliminates that state in favour of volatility, every Euler substep requires a recovered square root to remain real. The time-varying risk-price specification defines exchange-rate volatility through an implicit quadratic identity whose leading coefficient is negative at the reported U.S.–U.K. estimates, so that system admits either zero or two positive volatility branches and never exactly one. The full Brownian correlation matrix requires a joint positive-semidefinite constraint, not merely pairwise correlations in $[-1, 1]$. Finally, the minimum-incompleteness rule is a linear programme whose solution is always a corner, which we verify against the published tables, and together with the binding nonnegativity constraint it yields a hybrid and potentially nonregular estimator not covered by the unconstrained SML theorem.

The aim is not to relitigate the substantive economics of a paper published more than two decades ago. Nor do we infer from a false theorem that every numerical estimate obtained with finite M and S must be poor. The aim is narrower and mathematical: to identify the correct dimensional scale of the endpoint estimator, separate sequential from joint convergence, and determine what the published asymptotic arguments do and do not establish.

The remainder is organised as follows. Section 2 defines the estimator. Section 3 gives exact Brownian results and the counterexample. Section 4 derives the general diffusion moment law and corrected CLT. Section 5 studies the bias–variance trade-off. Section 6 discusses log likelihood and parameter estimation. Section 7 analyses the published proof. Section 8 studies the exchange-rate diffusion. Section 9 presents reproducible numerical evidence, and Section 10 discusses variance-reduction alternatives.

2 The endpoint simulated-likelihood estimator

Let $Y_t \in \mathbb{R}^K$ satisfy the stochastic differential equation

$$dY_t = \mu(Y_t, t; \theta)\, dt + \Sigma(Y_t, t; \theta)\, dW_t, \qquad V = \Sigma\Sigma^{\mathsf{T}}, \tag{1}$$

where W_t is a Brownian motion of suitable dimension and $\theta \in \boldsymbol{\Theta} \subset \mathbb{R}^L$. Coefficients are assumed regular enough for a unique strong solution in the sense of Karatzas and Shreve (1991). We suppress t and θ where no ambiguity arises. Consider one observation interval of length one, from $Y_0 = x$ to $Y_1 = y$. The normalisation to unit length is only notational.

For $M \geq 2$, set $h = 1/M$ and define the Euler chain (Kloeden and Platen, 1992)

$$Y_{m+1}^M = Y_m^M + \mu(Y_m^M, mh)h + \Sigma(Y_m^M, mh)\sqrt{h}\,\varepsilon_{m+1}, \qquad m = 0, \dots, M-1, \tag{2}$$

with $Y_0^M = x$ and independent $\varepsilon_m \sim \mathcal{N}(0, I)$. Conditional on $Y_m^M = z$, the one-step density is

$$g_h(y \mid z) = \varphi_K(y; z + \mu(z, 1-h)h, hV(z, 1-h))\,. \tag{3}$$

Let $Z_h = Y_{M-1}^M$ denote the preterminal Euler state and let $r_h(z \mid x)$ be its density. The M-step Euler transition density is

$$q_M(y \mid x) = \mathbb{E}[g_h(y \mid Z_h)] = \int_{\mathbb{R}^K} g_h(y \mid z) r_h(z \mid x)\, dz. \tag{4}$$

The endpoint simulation estimator is

$$\widehat{q}_{M,S}(y \mid x) = \frac{1}{S} \sum_{s=1}^{S} g_h(y \mid Z_{h,s}), \tag{5}$$

where $Z_{h,1}, \dots, Z_{h,S}$ are independent draws from the preterminal Euler law. This is equation (8) of Brandt and Santa-Clara (2002), with notation adapted to the present analysis.

For each fixed M, the strong law yields

$$\widehat{q}_{M,S}(y \mid x) \longrightarrow q_M(y \mid x) \quad \text{almost surely as } S \to \infty,$$

provided the summand is integrable. Under regularity conditions such as those of Bally and Talay (1995) and Bally and Talay (1996),

$$q_M(y \mid x) - p(y \mid x) = O(M^{-1}),$$

where p is the diffusion transition density. These two facts establish a sequential limit: first $S \to \infty$ for fixed M, then $M \to \infty$. They do not establish convergence along arbitrary joint sequences $M, S \to \infty$.

The geometry of (3) explains why. Its covariance is $hV(z)$. Under nondegeneracy, it is concentrated in a ball of radius $O(\sqrt{h})$ around the endpoint and has maximum height $O(h^{-K/2})$. The integral in (4) remains finite because the effective volume is $O(h^{K/2})$. A Monte Carlo average resolves that integral only if enough preterminal draws enter this shrinking region.

Definition 2.1 (Effective endpoint simulation size)**.** For a K-dimensional endpoint estimator with Euler step $h = 1/M$, define

$$R_{M,S} = Sh^{K/2} = \frac{S}{M^{K/2}}. \tag{6}$$

This is the expected-order number of simulations falling in an $O(\sqrt{h})$ neighbourhood of an endpoint with a regular positive preterminal density.

The next sections make this interpretation exact.

3 Exact Brownian analysis

Consider standard Brownian motion in $\mathbb{R}^K$,

$$dY_t = dW_t, \tag{7}$$

observed over a unit interval. Euler discretisation is exact. This benchmark is not an adversarial construction outside the scope of the published theory: with $\mu \equiv 0$ and $\Sigma \equiv I_K$, all coefficient derivatives vanish and are therefore continuous and bounded of every order, so Assumption 1 of Brandt and Santa-Clara (2002) holds, and $\Sigma\Sigma^\mathsf{T} = I_K$ is positive definite, so Assumption 2 holds. Lemmas 1–3 of that paper are stated under exactly these two assumptions. Any failure exhibited here is therefore a failure inside the stated hypotheses, and cannot be repaired by strengthening the regularity conditions. Set $Y_0 = x$ and evaluate the transition density at $Y_1 = y$. With $h = 1/M$,

$$Z_h \sim \mathcal{N}(x, (1-h)I_K)$$

and a single endpoint summand is

$$G_h = (2\pi h)^{-K/2} \exp\left[-\frac{\|y - Z_h\|^2}{2h}\right]. \tag{8}$$

3.1 Exact moments

Proposition 3.1 (Exact Brownian moments)**.** *Let $r > 0$. For the summand* (8)*,*

$$\begin{aligned}\mathbb{E}[G_h^r] &= (2\pi h)^{-rK/2}\left(1 + \frac{r(1-h)}{h}\right)^{-K/2} \exp\left[-\frac{r\|y-x\|^2}{2\{h + r(1-h)\}}\right] \\ &= (2\pi)^{-rK/2} h^{-(r-1)K/2}\{r - (r-1)h\}^{-K/2} \exp\left[-\frac{r\|y-x\|^2}{2\{r-(r-1)h\}}\right].\end{aligned} \tag{9}$$

In particular,

$$\mathbb{E}[G_h] = (2\pi)^{-K/2} \exp\left[-\frac{\|y-x\|^2}{2}\right] = p(y \mid x), \tag{10}$$

and

$$\mathbb{E}[G_h^2] = (2\pi)^{-K} h^{-K/2} (2-h)^{-K/2} \exp\left[-\frac{\|y-x\|^2}{2-h}\right]. \tag{11}$$

Proof. For $Z \sim \mathcal{N}(x, (1-h)I_K)$ and $c > 0$, the Gaussian identity

$$\mathbb{E}\left[e^{-c\|Z-y\|^2}\right] = \{1 + 2c(1-h)\}^{-K/2} \exp\left[-\frac{c\|x-y\|^2}{1+2c(1-h)}\right]$$

follows by completing the square. Apply it with $c = r/(2h)$ and multiply by the prefactor $(2\pi h)^{-rK/2}$. The cases $r = 1$ and $r = 2$ follow directly. □

Corollary 3.2 (Exact variance)**.** *For S independent simulations,*

$$\operatorname{Var}(\widehat{q}_{M,S}) = \frac{1}{S}\left[(2\pi)^{-K} h^{-K/2} (2-h)^{-K/2} e^{-\|y-x\|^2/(2-h)} - p(y \mid x)^2\right]. \tag{12}$$

Hence

$$\operatorname{Var}(\widehat{q}_{M,S}) \sim \frac{\tau_K^2(x,y)}{S h^{K/2}}, \qquad \tau_K^2(x,y) = (2\pi)^{-K} 2^{-K/2} e^{-\|y-x\|^2/2}. \tag{13}$$

The variance is not uniformly $O(1/S)$. It is $O(M^{K/2}/S)$. The estimator is unbiased in this example for every M, but its distribution becomes increasingly dominated by rare, very large contributions when $R_{M,S}$ is small.

3.2 A direct joint-limit counterexample

A variance calculation alone does not prove failure of convergence in probability because a non-uniformly integrable sequence can converge in probability while retaining large or divergent moments. The next theorem gives a direct failure result.

Theorem 3.3 (Collapse along $M = S$)**.** *Let $K > 2$, $x = y = 0$, and choose $M_n = S_n = n$. Then the Brownian endpoint estimator satisfies*

$$\widehat{q}_{n,n}(0 \mid 0) \longrightarrow 0 \qquad \text{in probability,} \tag{14}$$

although

$$p(0 \mid 0) = (2\pi)^{-K/2} > 0.$$

In every dimension the sequence satisfies the rate condition

$$\frac{\sqrt{S_n}}{M_n} = n^{-1/2} \longrightarrow 0$$

used in Theorems 1 and 2 of Brandt and Santa-Clara (2002), and it satisfies without any rate restriction the hypotheses of their Lemma 2, which requires only $M \to \infty$ and $S \to \infty$.

Proof. Write $h_n = 1/n$ and let $Z_{n,1}, \ldots, Z_{n,n}$ be the preterminal draws, each distributed as $\mathcal{N}(0, (1 - 1/n)I_K)$. Fix a constant $c > K$ and define

$$r_n = \sqrt{\frac{c \log n}{n}}.$$

Let A_n be the event that at least one draw lies in the ball $B(0, r_n)$. The Gaussian densities of the $Z_{n,s}$ are uniformly bounded for $n \geq 2$, so for a constant C_K,

$$\mathbb{P}(A_n) \leq n\mathbb{P}(\|Z_{n,1}\| \leq r_n) \leq C_K n r_n^K = C_K c^{K/2} n^{1-K/2} (\log n)^{K/2} \longrightarrow 0$$

because $K > 2$.

On A_n^c, every summand is bounded by

$$\left(\frac{n}{2\pi}\right)^{K/2} \exp\left(-\frac{n r_n^2}{2}\right) = (2\pi)^{-K/2} n^{(K-c)/2},$$

which tends to zero because $c > K$. The sample average obeys the same bound. Therefore, for every $\varepsilon > 0$,

$$\mathbb{P}(\widehat{q}_{n,n} > \varepsilon) \leq \mathbb{P}(A_n) + \mathbf{1}\left\{(2\pi)^{-K/2} n^{(K-c)/2} > \varepsilon\right\} \longrightarrow 0.$$

□

Remark 3.4 (How an unbiased estimator converges to zero)**.** For every n, $\mathbb{E}[\widehat{q}_{n,n}] = p(0 \mid 0)$. The convergence in probability to zero is sustained by increasingly rare simulations with increasingly large endpoint densities. The sequence is not uniformly integrable. This is precisely the rare-event geometry hidden by a fixed-M application of the strong law.

Corollary 3.5 (Contradiction in four dimensions)**.** *For $K = 4$, mean-square consistency requires*

$$\frac{S}{M^2} \longrightarrow \infty,$$

whereas the published condition $\sqrt{S}/M \to 0$ is equivalent to

$$\frac{S}{M^2} \longrightarrow 0.$$

The two conditions are mutually exclusive. Moreover, Theorem 3.3 shows that the published condition permits a sequence along which the density estimator converges to the wrong value.

Remark 3.6 (Where the conflict is unconditional)**.** The incompatibility in Corollary 3.5 is a property of dimensions $K \geq 4$. Mean-square consistency requires $S/M^{K/2} \to \infty$ and the published condition requires $S/M^2 \to 0$; for $K \geq 4$ the first demands $S \gtrsim M^{K/2} \geq M^2$ and the second demands $S \ll M^2$, so no sequence satisfies both. For $K \leq 3$ the two are compatible: with $K = 3$ and $S = M^{7/4}$ one has $S/M^{3/2} \to \infty$ and $\sqrt{S}/M \to 0$ simultaneously. Theorem 3.3 is nevertheless not confined to $K \geq 4$. It shows that even when the published condition admits consistent sequences, as at $K = 3$, it also admits inconsistent ones, so the condition cannot be what makes the theorem work. The empirical application of Brandt and Santa-Clara (2002) has $K = 4$, where no admissible sequence exists at all.

3.3 Correct triangular-array central limit theorem

Theorem 3.7 (Brownian endpoint CLT)**.** *Let $M \to \infty$ and $S \to \infty$ such that*

$$Sh^{K/2} = \frac{S}{M^{K/2}} \longrightarrow \infty. \tag{15}$$

Then

$$\sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S}(y \mid x) - p(y \mid x)\} \Longrightarrow \mathcal{N}\Big(0, \tau_K^2(x,y)\Big), \tag{16}$$

where $\tau_K^2(x, y)$ is defined in (13)*.*

Proof. Within each row, the G_h are independent and identically distributed. By Proposition 3.1,

$$\operatorname{Var}(G_h) \sim \tau_K^2 h^{-K/2}, \qquad \mathbb{E}[|G_h|^3] = O(h^{-K}).$$

The same order holds for the third absolute centred moment. The Lyapunov ratio is therefore

$$\frac{S\mathbb{E}|G_h - \mathbb{E}G_h|^3}{\{S \operatorname{Var}(G_h)\}^{3/2}} = O\Big(\frac{1}{\sqrt{Sh^{K/2}}}\Big) \longrightarrow 0.$$

The triangular-array Lyapunov central limit theorem applies. The variance normalisation follows from (13), and the Euler bias is zero for Brownian motion. □

The correct normalisation depends on dimension. A fixed-M $\sqrt{S}$-CLT is valid, but it is not stable as $M \to \infty$.

4 General multidimensional diffusions

The Brownian calculation reflects a local property of Gaussian endpoint kernels. We now derive the generic moment law. The assumptions below are deliberately stronger than necessary so that the mechanism is transparent.

Let $G_h = g_h(y \mid Z_h)$ be one summand in (5). Fix x and y.

Assumption 4.1 (Local regularity and domination)**.** *As $h \downarrow 0$:*

(i) *$\mu(z)$ and $V(z)$ are continuous in a neighbourhood of y, and $V(y)$ is positive definite;*
(ii) *on that neighbourhood, the eigenvalues of $V(z)$ are bounded between constants $0 < \underline{v} < \overline{v} < \infty$ and μ is bounded;*
(iii) *the preterminal Euler density $r_h(z \mid x)$, which is the density of the Euler chain at time $1 - h$ rather than at time 1, converges as $h \downarrow 0$ locally uniformly at y to a continuous density $p(z \mid x)$, with $p(y \mid x) > 0$. This is a joint requirement in time and space: it combines convergence of the Euler law to the diffusion law with continuity of $t \mapsto p_t(\cdot \mid x)$ at $t = 1$ in the locally uniform topology near y;*

(iv) *the family $r_h(\cdot \mid x)$ is uniformly bounded and the contribution to the moment integral from outside any fixed neighbourhood of y is dominated by an integrable Gaussian envelope after multiplication by the appropriate power of h.*

Uniform ellipticity, bounded smooth coefficients, and standard Gaussian density bounds imply versions of these conditions. They are compatible with the regular setting used to justify Euler density expansions in Bally and Talay (1995) and Bally and Talay (1996). They are not compatible with the square-root boundaries in the later financial application; that distinction is discussed in Section 8.

Theorem 4.2 (Local moment law)**.** *Under Assumption 4.1, for every fixed $r > 1$,*

$$h^{(r-1)K/2}\mathbb{E}[G_h^r] \longrightarrow c_{r,K}\, p(y \mid x)\, |V(y)|^{-(r-1)/2}, \tag{17}$$

where

$$c_{r,K} = (2\pi)^{-(r-1)K/2} r^{-K/2}. \tag{18}$$

In particular,

$$h^{K/2}\mathbb{E}[G_h^2] \longrightarrow (4\pi)^{-K/2} p(y \mid x)|V(y)|^{-1/2}. \tag{19}$$

Proof. Write $z = y + \sqrt{h}\,u$. On a fixed neighbourhood of y,

$$\begin{aligned} G_h^r &= (2\pi h)^{-rK/2}|V(z)|^{-r/2} \\ &\quad \times \exp\left[-\frac{r}{2h}\{y - z - h\mu(z)\}^\top V(z)^{-1}\{y - z - h\mu(z)\}\right]. \end{aligned}$$

Since $dz = h^{K/2}du$, multiplication by $h^{(r-1)K/2}$ cancels the powers of h. Pointwise in u,

$$V(y + \sqrt{h}\,u) \to V(y), \qquad r_h(y + \sqrt{h}\,u \mid x) \to p(y \mid x),$$

and the exponent converges to

$$-\frac{r}{2}u^\top V(y)^{-1}u.$$

The local eigenvalue bounds give an integrable Gaussian envelope. Dominated convergence and the tail condition yield

$$\begin{aligned} &\lim_{h\downarrow 0} h^{(r-1)K/2}\mathbb{E}[G_h^r] \\ &\quad = (2\pi)^{-rK/2}|V(y)|^{-r/2}p(y \mid x) \int_{\mathbb{R}^K} \exp\left[-\frac{r}{2}u^\top V(y)^{-1}u\right] du \\ &\quad = (2\pi)^{-rK/2}|V(y)|^{-r/2}p(y \mid x)(2\pi/r)^{K/2}|V(y)|^{1/2}, \end{aligned}$$

which is (17). □

Corollary 4.3 (Generic variance scale)**.** *Let $q_M(y \mid x) = \mathbb{E}[G_h]$ and suppose $q_M(y \mid x) \to p(y \mid x)$. Then*

$$\operatorname{Var}(\widehat{q}_{M,S}) \sim \frac{\tau^2(x,y)}{Sh^{K/2}}, \qquad \tau^2(x,y) = (4\pi)^{-K/2}p(y \mid x)|V(y)|^{-1/2}. \tag{20}$$

Theorem 4.4 (Generic endpoint CLT)**.** *Under Assumption 4.1, if $Sh^{K/2} \to \infty$, then*

$$\sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S}(y \mid x) - q_M(y \mid x)\} \Longrightarrow \mathcal{N}(0, \tau^2(x,y)). \tag{21}$$

Proof. The $r = 2$ and $r = 3$ cases of Theorem 4.2 imply

$$\operatorname{Var}(G_h) \sim \tau^2 h^{-K/2}, \qquad \mathbb{E}|G_h - \mathbb{E}G_h|^3 = O(h^{-K}).$$

The Lyapunov ratio is again $O\{(Sh^{K/2})^{-1/2}\}$. □

Corollary 4.5 (CLT centred at the diffusion density)**.** *Suppose in addition that*

$$q_M(y \mid x) - p(y \mid x) = b(x,y)h + o(h). \tag{22}$$

If

$$Sh^{K/2} \to \infty, \qquad \sqrt{Sh^{K/2}}\, h \to 0, \tag{23}$$

then

$$\sqrt{Sh^{K/2}}\,\{\widehat{q}_{M,S}(y \mid x) - p(y \mid x)\} \Longrightarrow \mathcal{N}(0, \tau^2(x,y)). \tag{24}$$

In terms of M and S, the second condition is

$$\frac{\sqrt{S}}{M^{1+K/4}} \longrightarrow 0. \tag{25}$$

For $K = 4$, a feasible regime is

$$M^2 \ll S \ll M^4. \tag{26}$$

Remark 4.6 (The dimensional conflict)**.** The condition $\sqrt{S}/M \to 0$ suppresses an $O(M^{-1})$ Euler bias under an incorrectly fixed Monte Carlo scale. Once the variance inflation $M^{K/2}$ is included, the bias must be compared with the standard deviation $M^{K/4}/\sqrt{S}$, producing (25). In four dimensions, the lower consistency requirement and the upper bias requirement leave the nonempty interval (26). The published condition places S below the lower boundary of that interval.

5 Bias–variance trade-off and dimensional rate

Suppose the Euler density admits (22). Because (5) is unbiased for q_M,

$$\begin{aligned}\mathbb{E}\Big[(\widehat{q}_{M,S} - p)^2\Big] &= (q_M - p)^2 + \mathrm{Var}(\widehat{q}_{M,S}) \\ &= b(x,y)^2 h^2 + \frac{\tau^2(x,y)}{Sh^{K/2}} + o\Big(h^2 + \frac{1}{Sh^{K/2}}\Big).\end{aligned} \tag{27}$$

Proposition 5.1 (Pointwise optimal discretisation)**.** *If $b(x,y) \neq 0$, the leading terms in* (27) *are balanced by*

$$h \asymp S^{-2/(K+4)}, \qquad M \asymp S^{2/(K+4)}. \tag{28}$$

The resulting optimal pointwise mean squared error is

$$\mathrm{MSE}_{\mathrm{opt}} \asymp S^{-4/(K+4)}. \tag{29}$$

Proof. Differentiate the leading expression $Bh^2 + C/(Sh^{K/2})$ with respect to h or equate its two terms. This gives $h^{K/2+2} \asymp S^{-1}$ and hence (28). Substitution yields (29). □

The rate is the familiar structure of a kernel-density problem: reducing the bandwidth controls approximation bias but decreases the effective number of simulations. The dimension enters through the volume of the final Gaussian kernel. Some illustrative exponents are

Dimension K	M_{opt}	Optimal MSE
1	$S^{2/5}$	$S^{-4/5}$
2	$S^{1/3}$	$S^{-2/3}$
4	$S^{1/4}$	$S^{-1/2}$
8	$S^{1/6}$	$S^{-1/3}$

Table 1: Principal endpoint-density rate conditions.

Property	General dimension K	Four dimensions
Monte Carlo mean-square consistency	$S/M^{K/2} \to \infty$	$S/M^2 \to \infty$
Density CLT centred at q_M	$S/M^{K/2} \to \infty$	$S/M^2 \to \infty$
Euler bias negligible in corrected CLT	$\sqrt{S}/M^{1+K/4} \to 0$	$S/M^4 \to 0$
Condition in Brandt and Santa-Clara (2002)	$\sqrt{S}/M \to 0$	$S/M^2 \to 0$

This does not mean that M should always be chosen by pointwise MSE. Likelihood estimation requires accuracy over many endpoints and parameter values, and higher-order or bridge-based methods can change both constants and rates. It does mean that increasing M while holding S fixed, or while letting S grow too slowly relative to dimension, can make the simulation approximation worse rather than better.

6 From density error to log-likelihood error

For observations $Y_0, \dots, Y_N$, the simulated log likelihood is a sum of terms

$$\log \widehat{q}_{M,S}(Y_{n+1} \mid Y_n; \theta).$$

Even when $\widehat{q}_{M,S}$ is unbiased for q_M, its logarithm is not unbiased. The effective endpoint size $R_{M,S} = Sh^{K/2}$ controls this nonlinearity.

Proposition 6.1 (Log-density delta expansion)**.** *Fix an endpoint with $q_M \to p > 0$ and suppose the conditions of Theorem 4.4 hold. If $R_{M,S} = Sh^{K/2} \to \infty$, then*

$$\sqrt{R_{M,S}}\,\{\log \widehat{q}_{M,S} - \log q_M\} \Longrightarrow \mathcal{N}\left(0, \frac{\tau^2}{p^2}\right), \tag{30}$$

and

$$\log \widehat{q}_{M,S} - \log q_M = \frac{\widehat{q}_{M,S} - q_M}{q_M} - \frac{(\widehat{q}_{M,S} - q_M)^2}{2q_M^2} + o_{\mathbb{P}}(R_{M,S}^{-1}). \tag{31}$$

Whenever the second-order expansion is uniformly integrable, it follows that

$$\mathbb{E}[\log \widehat{q}_{M,S}] - \log q_M = -\frac{\tau^2}{2p^2 R_{M,S}} + o(R_{M,S}^{-1}). \tag{32}$$

Proof. The first statement is the delta method applied to Theorem 4.4. Since $(\widehat{q} - q_M)/q_M = O_{\mathbb{P}}(R^{-1/2})$ and q_M is bounded away from zero, Taylor expansion of $\log(1+u)$ gives (31). Taking expectations requires uniform integrability of the remainder and lower-tail control; under that additional condition, the first-order term has zero expectation and the second-order term gives (32). □

The leading nonlinear scale is

$$R_{M,S}^{-1} = \frac{M^{K/2}}{S}, \tag{33}$$

not uniformly $1/S$. The statement in Brandt and Santa-Clara (2002, p. 171) that the logarithmic transformation induces a bias of order $1/S$ is valid with M fixed. It is not a joint-M, S order.

Suppose simulations are independent across observations and the local constants are uniformly bounded. For the average simulated criterion

$$\widehat{Q}_{N,M,S}(\theta) = \frac{1}{N}\sum_{n=0}^{N-1} \log \widehat{q}_{n,M,S}(\theta),$$

the stochastic average of the first-order simulation error has standard deviation of order

$$(NR_{M,S})^{-1/2},$$

while the average nonlinear bias remains of order $R_{M,S}^{-1}$. The simulation bias does not average out across observations.

A full estimator theorem requires uniform control in θ of densities, scores, and Hessians. The following standard perturbation statement separates those requirements from the endpoint calculations.

Proposition 6.2 (Abstract equivalence to exact MLE)**.** *Let $\widehat{\theta}_N$ maximise an exact average log likelihood Q_N, and let $\widehat{\theta}_{N,M,S}$ maximise a simulated criterion $\widehat{Q}_{N,M,S}$. Assume:*

(i) $\widehat{\theta}_N \to \theta_0$ and $\sqrt{N}(\widehat{\theta}_N - \theta_0)$ has the usual exact-MLE limit;

(ii) on a neighbourhood of θ_0,

$$\sup_\theta \|\nabla\widehat{Q}_{N,M,S}(\theta) - \nabla Q_N(\theta)\| = o_{\mathbb{P}}(N^{-1/2});$$

(iii) on the same neighbourhood,

$$\sup_\theta \|\nabla^2\widehat{Q}_{N,M,S}(\theta) - \nabla^2 Q_N(\theta)\| = o_{\mathbb{P}}(1);$$

(iv) the exact criterion has a nonsingular limiting Hessian.

Then

$$\sqrt{N}(\widehat{\theta}_{N,M,S} - \widehat{\theta}_N) \to 0 \quad \textit{in probability.}$$

Proof. Expand the simulated first-order condition around $\widehat{\theta}_N$ and use the uniform score and Hessian bounds together with nonsingularity of the limiting exact Hessian. □

Under uniform analogues of the pointwise density expansion, plausible minimum requirements for first-order equivalence include

$$Sh^{K/2} \to \infty, \qquad \frac{\sqrt{N}}{Sh^{K/2}} \to 0, \qquad \sqrt{N}\,h \to 0, \tag{34}$$

corresponding respectively to vanishing score simulation noise, vanishing nonlinear simulation bias at the $N^{-1/2}$ scale, and negligible Euler bias. Derivatives of the simulated paths and Gaussian kernels can impose stronger conditions. Therefore (34) is a design implication, not a claimed complete theorem for arbitrary diffusions.

The broader Monte Carlo likelihood literature treats joint data and simulation limits through empirical-process, importance-sampling, or exact-simulation arguments rather than by combining two pointwise limits; see, among others, Lee (1995), Sung and Geyer (2007), Beskos, Papaspiliopoulos, and Roberts (2009), and Miasojedow et al. (2016).

7 What fails in the published asymptotic argument

This section distinguishes statements refuted by the Brownian example from steps that are merely unproved under the stated assumptions.

7.1 Lemma 2: sequential convergence is used as joint convergence

The proof of Lemma 2 in Brandt and Santa-Clara (2002) applies the strong law to the Monte Carlo average and then invokes convergence of the Euler density. For every fixed M,

$$\widehat{q}_{M,S} \to q_M \quad \text{as } S \to \infty,$$

and separately

$$q_M \to p \quad \text{as } M \to \infty.$$

This gives the iterated limit

$$\lim_{M\to\infty} \lim_{S\to\infty} \widehat{q}_{M,S} = p.$$

It does not give convergence along every sequence $(M_n, S_n) \to (\infty, \infty)$. The summand changes with M and its variance diverges as $M^{K/2}$. Theorem 3.3 directly refutes the stated joint conclusion.

Two features of Lemma 2 are worth isolating. It is stated as a joint limit in M and S with no rate restriction whatsoever, so the refutation does not depend on the separate discussion of $\sqrt{S}/M \to 0$ below; and it is stated under Assumptions 1 and 2 alone, which standard Brownian motion satisfies. The counterexample therefore meets the lemma on its own terms.

7.2 Lemma 3: the fixed-M CLT is not uniform in M

The paper uses a $\sqrt{S}$ central limit theorem and requires $\sqrt{S}/M \to 0$ to suppress the Euler bias. The variance of the triangular-array summand is not asymptotically constant. It is of order $M^{K/2}$. The correct standardisation is $\sqrt{S/M^{K/2}}$. The Euler bias must then be compared with that scale, producing (25).

7.3 Lemma 4: an $O_{\mathbb{P}}$ term is treated as $o_{\mathbb{P}}$

Let $x_n = \widehat{q}_n - p_n$. A nondegenerate central limit theorem implies $x_n = O_{\mathbb{P}}(S^{-1/2})$ for fixed M, not $o_{\mathbb{P}}(S^{-1/2})$. The Taylor expansion

$$\log\left(1 + \frac{x_n}{p_n}\right) = \frac{x_n}{p_n} + O_{\mathbb{P}}(x_n^2)$$

retains a first-order stochastic term. In the joint limit, the correct order is $O_{\mathbb{P}}\{(Sh^{K/2})^{-1/2}\}$. Moreover, summing pointwise errors over N observations requires lower bounds or tail control for p_n, dependence control, and a triangular-array argument. These are not supplied.

7.4 Lemma 5: convergence of objective values does not control maximisers

An argmax result requires uniform convergence of the criterion and separation or local curvature around the maximiser. A bounded Hessian gives an upper quadratic bound. To infer that two maximisers are close from a small objective difference, one needs a lower quadratic bound or a negative Hessian whose smallest eigenvalue is bounded away from zero after normalisation. Pointwise convergence and identification alone are insufficient.

The structure of the argument makes this explicit. Summing the two second-order expansions leaves a difference of objective values on the left and a term proportional to $(\widehat{\theta}_N - \widehat{\theta}_{N,M,S})^2$ on the right, whose coefficient is an average of two Hessians. Assumption 4 of Brandt and Santa-Clara (2002) bounds that coefficient above. Dividing a small left-hand side by a coefficient that is only bounded above yields no bound at all on the squared distance; what is required is a bound below, that is, a uniformly negative definite Hessian in a neighbourhood. The conclusion drawn, $\widehat{\theta}_{N,M,S} - \widehat{\theta}_N = o(N^{1/2}/S^{1/4})$, also inherits whatever is wrong with the $o(N/S^{1/2})$ input from Lemma 4.

7.5 Lemmas 6 and 7: exact-MLE asymptotics require additional structure

Consistency of the exact diffusion MLE ordinarily requires a uniform law of large numbers, global or local identification, and stability assumptions such as stationarity and ergodicity when $N \to \infty$. A score expansion around θ_0 is not by itself a proof of consistency because the intermediate Hessian is evaluated between the estimator and θ_0.

Similarly, the martingale central limit theorem invoked from Hall and Heyde (1980) requires convergence of the realised conditional quadratic variation and a conditional Lindeberg condition. Defining Fisher information as an expectation does not make the realised conditional variance condition automatic. Boundedness of $\partial p/\partial\theta$ is also not a bound on the score $(\partial p/\partial\theta)/p$ when p can be small; strict positivity of p at each point is weaker than the uniform lower bound the argument needs.

These points do not imply that no correct exact-MLE theorem can be written. They imply that the short proof in Appendix A does not establish the stated result, even apart from the false transition-density joint limit.

8 Mathematical status of the exchange-rate application

The empirical application estimates a four-dimensional system consisting of two interest rates, the log exchange rate, and exchange-rate volatility. It also reconstructs a market-incompleteness state. Several mathematical features are not covered by the generic SML theorem.

8.1 The implemented design and the paper's own rate conditions

Before turning to the structure of the model it is worth recording what the asymptotic theorems require of N, M, and S, and what the implementation uses. Theorem 2 of Brandt and Santa-Clara (2002) imposes two conditions jointly,

$$\frac{\sqrt{S}}{M} \longrightarrow 0, \qquad \frac{N}{S^{1/4}} \longrightarrow 0, \tag{35}$$

the second entering through the $o(N^{1/2}/S^{1/4})$ bound of their Lemma 5. Together these require

$$N^4 \ll S \ll M^2. \tag{36}$$

The estimation uses $N = 544$ weekly observations from January 1990 to May 2000, $M = 10$ Euler substeps, and $S = 5{,}000$ simulations together with 5,000 antithetic variates. At these values

$$\frac{\sqrt{S}}{M} = 7.07, \qquad \frac{N}{S^{1/4}} = 64.7,$$

and counting the antithetic draws as if they were independent gives 10.0 and 54.4. Neither ratio is small. To satisfy (36) at $N = 544$ one would need $S \gg N^4 \approx 8.8 \times 10^{10}$ and then $M \gg \sqrt{S} \gtrsim 3 \times 10^5$, which exceeds the implemented values by roughly seven and four orders of magnitude respectively.

This observation is logically independent of everything else in this paper. Even granting the published lemmas in full, the reported estimates are not covered by the asymptotic theory that is invoked to justify them, because the design does not approach the region in which that theory operates. Corollary 3.5 then shows that in $K = 4$ no design could have approached it, since (35) and mean-square consistency of the transition density cannot hold together.

The dimensional diagnostic of Definition 2.1 gives the complementary finite-sample reading. With $K = 4$, $M = 10$, and $S = 5{,}000$,

$$R_{M,S} = \frac{S}{M^{K/2}} = \frac{5{,}000}{100} = 50, \tag{37}$$

or 100 if the antithetic variates are counted as independent, which they are not. Each transition density in a 544-term log likelihood is therefore resolved by an effective sample of order 10^2, not 10^4. Nothing here shows that such a design is inaccurate; the point is that its accuracy is governed by (37) rather than by S, and that doubling M in the final round of optimisation, as the paper reports doing, divides $R_{M,S}$ by four while appearing to refine the approximation.

8.2 Square-root coefficients violate the generic assumptions

The interest rates are modelled as correlated Cox–Ingersoll–Ross processes (Cox et al., 1985),

$$dr_t = \lambda(\vartheta - r_t)dt + \sigma\sqrt{r_t}\,dW_t, \qquad dr_t^* = \lambda^*(\vartheta^* - r_t^*)dt + \sigma^*\sqrt{r_t^*}\,dW_t^*. \tag{38}$$

The generic theory assumes infinitely differentiable coefficients with bounded derivatives and a positive-definite covariance matrix. The function $r \mapsto \sqrt{r}$ is not differentiable at zero, has unbounded derivatives near zero, is not defined on the negative half-line as a real coefficient, and makes the covariance degenerate at the boundary. The same singularity appears explicitly in the transformed system of Appendix B of Brandt and Santa-Clara (2002), whose drift for $\sqrt{r_t}$ contains the term $-\frac{1}{8}\sigma^2 r_t^{-1/2}$.

It is important to separate this structural violation from its practical consequences, and the two point in opposite directions for the interest rates. Table 2 reports the Feller ratio $2\lambda\vartheta/\sigma^2$ at the estimates of Table 3 of Brandt and Santa-Clara (2002). It ranges from 6.4 to 48.5, so in every case the ratio comfortably exceeds one, the origin is unattainable for the exact processes, and the interest-rate states are far from the boundary. The corresponding Euler negativity probabilities are numerically nil: with $h = 1/520$, corresponding to weekly data and $M = 10$, a step taken from the German long-run mean $\vartheta^* = 0.064$ has mean-to-standard-deviation ratio 137, and even a step taken from the implausibly low state $r^* = 0.001$ has ratio 17.4. The assumptions of the generic theorem are genuinely violated by (38), and the estimator applied is therefore not the estimator analysed; but for r and r^* that gap is a matter of proof coverage rather than of numerical damage.

The market-incompleteness state behaves differently. Let $X_t = \epsilon_t^2$. Equation (37) of Brandt and Santa-Clara (2002) specifies

$$dX_t = (\beta^2 - 2\alpha X_t)dt + 2\beta\sqrt{X_t}\,dY_t, \tag{39}$$

which is a CIR process with $\kappa = 2\alpha$, long-run mean $\beta^2/(2\alpha)$, and diffusion coefficient 2β. Its Feller ratio is

$$\frac{2\kappa\vartheta_X}{\sigma_X^2} = \frac{2\beta^2}{4\beta^2} = \frac{1}{2}$$

for every $\alpha > 0$ and every $\beta \neq 0$. The ratio is exactly one half independently of the parameter values, so no estimate can move this process away from its boundary: zero is attainable, and the instantaneous covariance degenerates there. This is a structural feature of the specification, not a property of a particular estimate. Proposition 8.2 below shows that the boundary is not merely attainable in principle but attained exactly in the reported results.

Table 2: Feller ratios at the estimates reported in Table 3 of Brandt and Santa-Clara (2002). For a process $dZ = \kappa(\vartheta_Z - Z)dt + \sigma_Z\sqrt{Z}\,dB$ the ratio is $2\kappa\vartheta_Z/\sigma_Z^2$, and the origin is unattainable if and only if it is at least one. For the incompleteness state the ratio equals 1/2 for all admissible parameters.

System	State	κ	ϑ_Z	σ_Z	$2\kappa\vartheta_Z/\sigma_Z^2$
U.S. vs. U.K.	r (U.S.)	0.284	0.053	0.028	38.4
	r^* (U.K.)	0.486	0.074	0.056	22.9
	ϵ^2	0.640	0.0121	0.176	0.5
U.S. vs. Germany	r (U.S.)	0.305	0.058	0.027	48.5
	r^* (Germany)	0.088	0.064	0.042	6.4
	ϵ^2	0.676	0.0151	0.202	0.5

8.3 The simulated state and the reality of the incompleteness process

The previous subsection treats (39) as a state in its own right, which is how the model presents it. The estimator does not. Equation (31) and Appendix B of Brandt and Santa-Clara (2002) eliminate ϵ_t using the variance identity and simulate the four-dimensional state

$$(r_t,\ r_t^*,\ e_t,\ v_t), \tag{40}$$

where e_t is the log exchange rate and v_t is its instantaneous volatility, proxied by a one-week at-the-money implied volatility. The incompleteness state is therefore never discretised directly. This relocates the boundary problem rather than removing it, and the relocated version is the one that binds.

The identity that eliminates ϵ_t is

$$\epsilon_t = \sqrt{v_t^2 - (\phi_t^2 + \phi_t^{*2} - 2\rho_{ww^*}\phi_t\phi_t^*) - (\psi_t^2 + \psi_t^{*2} - 2\rho\,\psi_t\psi_t^*)}, \tag{41}$$

where $\phi_t = \phi\sqrt{r_t}$ and $\phi_t^* = \phi^*\sqrt{r_t^*}$ are the market prices of domestic and foreign interest-rate risk and ψ_t, ψ_t^* are the market prices of pure currency risk. The state ϵ_t appears in the diffusion of the log exchange rate, through $v_t\,dX_t = \phi_t dW_t - \phi_t^* dW_t^* + \psi_t dZ_t - \psi_t^* dZ_t^* + \epsilon_t dU_t$, and in the coefficients of the v_t equation obtained from Itô's lemma. Every Euler substep of (40) therefore requires the radicand of (41) to be nonnegative at the simulated state, not merely at the observed data. Ordinary Euler discretisation of a system whose coefficients are defined by (41) offers no such guarantee: the increments of v_t are conditionally Gaussian and unbounded below, while the subtracted terms are continuous and generically bounded away from zero.

This is the operative form of the degeneracy, and three features make it sharp rather than hypothetical. First, the constraint is not slack: Proposition 8.2 shows that the reported identification scheme drives $\min_t \epsilon_t^2$ to exactly zero, and Table 4 of Brandt and Santa-Clara (2002) reports a minimum excess volatility of 0.0000 in all six reported specifications. The simulated system is started, at least once in every sample, from a state exactly on the boundary. Second, the failure is not a recoverable numerical event such as a negative variance that can be floored at zero; it makes the drift and diffusion coefficients of the next substep complex-valued, so the chain is undefined rather than inaccurate. Third, the paper reports assigning zero likelihood to parameter values that violate $\epsilon_t^2 \geq 0$ at the observed dates, which is a restriction on parameters, not a rule for what the simulator does when an intermediate Euler state violates it.

As with the interest rates, a truncation, reflection, absorption, or exact-transition rule would resolve the issue, but each such rule defines a different approximating chain with a different transition density and requires its own convergence analysis. The published paper does not specify one. Figure 5 illustrates the mechanism on the incompleteness specification (39) taken at face value, where it can be computed in closed form: for an Euler step of size h the probability of a negative next state from $X_t = x > 0$ is

$$\Phi\left(-\frac{x + (\beta^2 - 2\alpha x)h}{2\beta\sqrt{xh}}\right). \tag{42}$$

The figure is an illustration of the class of failure, not a simulation of the implemented chain, whose boundary is (41) and whose exact crossing probability depends on the drift and diffusion of v_t derived in Appendix B.

8.4 The time-varying volatility specification is implicit

The time-varying pure-currency risk prices have the form

$$\psi_t = a_t + bv_t, \qquad \psi_t^* = a_t^* + b^* v_t, \tag{43}$$

where a_t and a_t^* depend on the interest-rate differential and log exchange rate. The variance identity is

$$v_t^2 = D_t + \psi_t^2 + (\psi_t^*)^2 - 2\rho\psi_t\psi_t^*, \tag{44}$$

where D_t collects the interest-rate risk term and ϵ_t^2.

Substituting (43) into (44) gives

$$(1 - \kappa_b)v_t^2 - 2\eta_t v_t - \zeta_t = 0, \tag{45}$$

where

$$\kappa_b = b^2 + (b^*)^2 - 2\rho bb^*, \tag{46}$$

$$\eta_t = a_t b + a_t^* b^* - \rho(a_t b^* + a_t^* b), \tag{47}$$

$$\zeta_t = D_t + a_t^2 + (a_t^*)^2 - 2\rho a_t a_t^*. \tag{48}$$

When the underlying two-dimensional correlation matrix is positive semidefinite and $D_t \geq 0$, one has $\zeta_t \geq 0$.

Proposition 8.1 (Branch structure of the implicit volatility)**.** *Let $\zeta_t > 0$.*

(i) If $\kappa_b < 1$, equation (45) *has exactly one positive root and one negative root, so the positive branch is unique.*

(ii) If $\kappa_b > 1$, equation (45) *has either no positive root or exactly two positive roots. A unique positive root is impossible except in the knife-edge case $\eta_t^2 = (\kappa_b - 1)\zeta_t$ of a repeated root. Specifically, positive roots exist if and only if $\eta_t < 0$ and $\eta_t^2 \geq (\kappa_b - 1)\zeta_t$, and they are then*

$$v_t^{\pm} = \frac{-\eta_t \pm \sqrt{\eta_t^2 - (\kappa_b - 1)\zeta_t}}{\kappa_b - 1}.$$

Proof. Rewrite (45) as $(\kappa_b - 1)v_t^2 + 2\eta_t v_t + \zeta_t = 0$.

(i) If $1 - \kappa_b > 0$ the product of the roots of (45) is $-\zeta_t/(1 - \kappa_b) < 0$, so the real roots have opposite signs. The discriminant is $4\{\eta_t^2 + (1 - \kappa_b)\zeta_t\} > 0$, hence exactly one root is positive.

(ii) If $\kappa_b - 1 > 0$ the product of the roots is $\zeta_t/(\kappa_b - 1) > 0$ and their sum is $-2\eta_t/(\kappa_b - 1)$. A positive product excludes roots of opposite signs and excludes a zero root, so the real roots, when they exist, are either both positive or both negative, and they are both positive precisely when the sum is positive, that is when $\eta_t < 0$. Real roots exist when the discriminant $4\{\eta_t^2 - (\kappa_b - 1)\zeta_t\}$ is nonnegative, and the stated formula is the quadratic formula. Equality of the two roots requires $\eta_t^2 = (\kappa_b - 1)\zeta_t$, a codimension-one condition on the state. □

Table 3: Quadratic coefficient implied by the model-C volatility loadings reported in Table 4 of Brandt and Santa-Clara (2002). Here $\kappa_b = b^2 + (b^*)^2 - 2\rho b b^*$.

Market	b	b^*	ρ	κ_b	$1-\kappa_b$
U.S. vs. U.K.	1.514	2.581	0.89	1.9982	−0.9982
U.S. vs. Germany	−0.142	0.127	0.94	0.0702	0.9298

Table 4 of Brandt and Santa-Clara (2002) reports, for model C,

$$(b, b^*, \rho) = (1.514, 2.581, 0.89)$$

for the U.S.–U.K. system. These values imply

$$\kappa_b \approx 1.9982, \qquad 1 - \kappa_b \approx -0.9982.$$

These estimates fall in case (ii) of Proposition 8.1, not merely outside case (i). For the U.S.–U.K. model-C specification the volatility identity therefore admits either no positive solution or two positive solutions at almost every state, and a unique positive volatility branch is not available. Which of the two branches is intended is not determined by the specification, and the branch structure is state dependent: as η_t crosses zero, or as $\eta_t^2 - (\kappa_b - 1)\zeta_t$ changes sign, the solution set changes between empty and a pair. Since $\zeta_t \geq 0$ requires only positive semidefiniteness of the two-dimensional correlation matrix and $D_t \geq 0$, both of which hold by construction, this is not a knife-edge consequence of a particular parameter estimate; it follows from $\kappa_b > 1$ alone. For the U.S.–Germany estimates $\kappa_b \approx 0.0702 < 1$ and case (i) applies, so that system does have a unique positive branch. The calculation is shown in Table 3.

This observation does not prevent the authors from evaluating (44) conditional on an observed implied volatility, which is what the estimation does at the sample dates. It matters when the equations are interpreted as a generative four-dimensional diffusion whose state v_t must exist uniquely and evolve continuously, which is what the simulation of (40) requires at every Euler substep. Appendix B defines the drift and diffusion of v_t only implicitly, with those coefficients appearing inside the formulas used to define them. An invertibility and branch selection condition is required.

8.5 Pairwise correlations do not guarantee a valid Brownian system

The application specifies an instantaneous correlation matrix for the four Brownian motions (W_t, W_t^*, X_t, Y_t). Write it in block form as

$$R_t = \begin{pmatrix} A_t & r_t \\ r_t^\top & 1 \end{pmatrix}, \tag{49}$$

where A_t is the 3×3 block for (W_t, W_t^*, X_t) and r_t contains the three correlations with Y_t. A valid Brownian system requires $R_t \succeq 0$ for every state and parameter value. If $A_t \succ 0$, the Schur-complement condition is

$$1 - r_t^\top A_t^{-1} r_t \geq 0. \tag{50}$$

Constraining each entry of r_t to lie in $[-1, 1]$ is not sufficient. The paper does not state a Cholesky, partial-correlation, hyperspherical, or Schur-complement parameterisation that enforces (50). It is possible that the numerical implementation rejected invalid matrices, but such a rule is not part of the mathematical specification.

8.6 Minimum incompleteness is an identifying selection rule

The exchange-rate dynamics do not separately identify one risk-premium component and the degree of market incompleteness. The paper resolves this by choosing the unidentified quantity, described as chosen "in a least-squares sense," to minimise $\sum_t \epsilon_t^2$ subject to $\epsilon_t^2 \geq 0$ at every date. Under constant pure-currency risk prices the free quantity is the product $\sigma_{z^*}\epsilon_{c^*}$; under time-varying prices it is the correlation ρ_{zz^*}.

The description as a least-squares problem is misleading in a way that determines the answer. Because ϵ_t^2 is itself the residual, the objective $\sum_t \epsilon_t^2$ is linear, not quadratic, in the free quantity, so the solution is always a corner of the feasible set rather than an interior least-squares projection.

Proposition 8.2 (Minimum-incompleteness solution)**.** *Let A_t denote the observed exchange-rate variance after subtracting all already identified components and let the free quantity enter linearly, so that $\epsilon_t^2 = A_t - \gamma_t c$ for known loadings γ_t and a scalar c.*

(i) Constant risk prices. Here $\gamma_t \equiv 1$ and $c = \sigma_z \epsilon_c + \sigma_{z^*}\epsilon_{c^*}$. The problem

$$\min_c \sum_{t=1}^{N} (A_t - c) \quad \textit{subject to} \quad A_t - c \geq 0 \quad (t = 1, \ldots, N) \tag{51}$$

has the unique solution

$$c^* = \min_{1 \leq t \leq N} A_t, \qquad \epsilon_t^2 = A_t - \min_s A_s. \tag{52}$$

(ii) Time-varying risk prices. Here $c = \rho_{zz^*}$ and $\gamma_t = -2\psi_t\psi_t^*$, so the objective is again linear in c with slope $-\sum_t \gamma_t$. Provided $\sum_t \gamma_t \neq 0$, the optimum is again attained at a corner of the feasible polyhedron $\{c : A_t - \gamma_t c \geq 0 \ \forall t\}$, at which at least one constraint is active.

In both cases $\min_t \epsilon_t^2 = 0$ exactly, unless the corner is attained only in a limiting or numerically constrained sense.

Proof. (i) The objective equals $\sum_t A_t - Nc$ and is strictly decreasing in c. Feasibility requires $c \leq \min_t A_t$. Hence the largest feasible value is the unique optimum, and the constraint indexed by $\arg\min_t A_t$ is active.

(ii) The objective equals $\sum_t A_t - c\sum_t \gamma_t$, which is affine in c with nonzero slope, and the feasible set is a closed interval determined by the constraints $A_t - \gamma_t c \geq 0$. An affine function with nonzero slope on a closed interval attains its minimum at an endpoint, at which the corresponding constraint holds with equality. □

The prediction is verifiable in the published output. Table 4 of Brandt and Santa-Clara (2002) reports the minimum of the inferred excess volatility as 0.0000 in every one of the six reported specifications, that is for models A, B, and C in both the U.S.–U.K. and the U.S.–Germany systems. Part (ii) of the proposition is what accounts for models B and C, where the free quantity is a correlation rather than a level, and where the paper separately reports that the constraint is binding.

The mechanism is acknowledged in part in the original paper, which observes on p. 193 that with constant risk prices the inferred excess volatility is essentially $\epsilon_t = \{v_t^2 - \min_s v_s^2\}^{1/2}$, and draws from this the conclusion that a volatile v_t leaves a large average residual. Proposition 8.2 formalises that observation, shows that it is a property of the optimisation rather than of the data, and extends it to the time-varying specifications where the same corner structure holds for a different free parameter. The consequence for interpretation is that the volatility of volatility, rather than its level, mechanically drives much of the estimated excess volatility:

the residual is the observed variance minus its sample minimum after other adjustments, so its average size is a measure of the dispersion of v_t^2 and is bounded below by construction.

The rule is transparent, but it is not identification by the likelihood of the original structural model. It selects a member of an observationally equivalent or weakly identified class by imposing the smallest feasible incompleteness. The paper describes it as a "prior of sorts." Mathematically, the final estimator combines simulated likelihood, auxiliary least squares from bond yields, a minimum-incompleteness corner rule, and inequality constraints. A further consequence, used in Section 8.3, is that the reconstructed incompleteness state sits exactly on its own degenerate boundary at the minimising date.

8.7 Binding constraints imply nonregular inference

The empirical optimisation assigns zero likelihood to parameter values for which any reconstructed ϵ_t^2 is negative. The paper later states that the constraint is binding in the time-varying specifications. A criterion with an active inequality boundary is not twice continuously differentiable across the boundary, and the usual unconstrained normal approximation need not hold. Boundary MLEs and likelihood-ratio statistics commonly have tangent-cone or mixture limits rather than the standard interior distributions; see Self and Liang (1987).

Stacking first-order conditions and reporting ordinary GMM standard errors does not automatically account for:

- a sample-dependent feasible set defined by N inequalities;
- changes in the active set;
- nondifferentiability of an inner constrained optimisation;
- structural parameters selected through minimum incompleteness;
- an optimum on or near the boundary.

A separate constrained-estimation analysis is needed before conventional symmetric standard errors and Wald tests can be justified.

9 Numerical evidence

All numerical results in this section are generated by the accompanying Python script. They use exact Gaussian benchmarks where possible and are intended to illustrate the theorems rather than replace them.

9.1 Second-moment scaling

Figure 1 plots

$$h^{K/2}\mathbb{E}[G_h^2]$$

divided by its limiting value for $K \in \{1, 2, 4, 8\}$. The exact formula (11) converges to the constant predicted by Theorem 4.2. The unscaled second moment grows as $M^{K/2}$.

9.2 Three joint asymptotic paths

For $K = 4$, Figure 2 compares the exact relative root mean squared error under $S = M$, $S = M^2$, and $S = M^3$. The first path satisfies the published condition but has increasing RMSE. The second keeps the effective endpoint size constant and does not become consistent. The third satisfies $S/M^2 \to \infty$ and has decreasing RMSE.

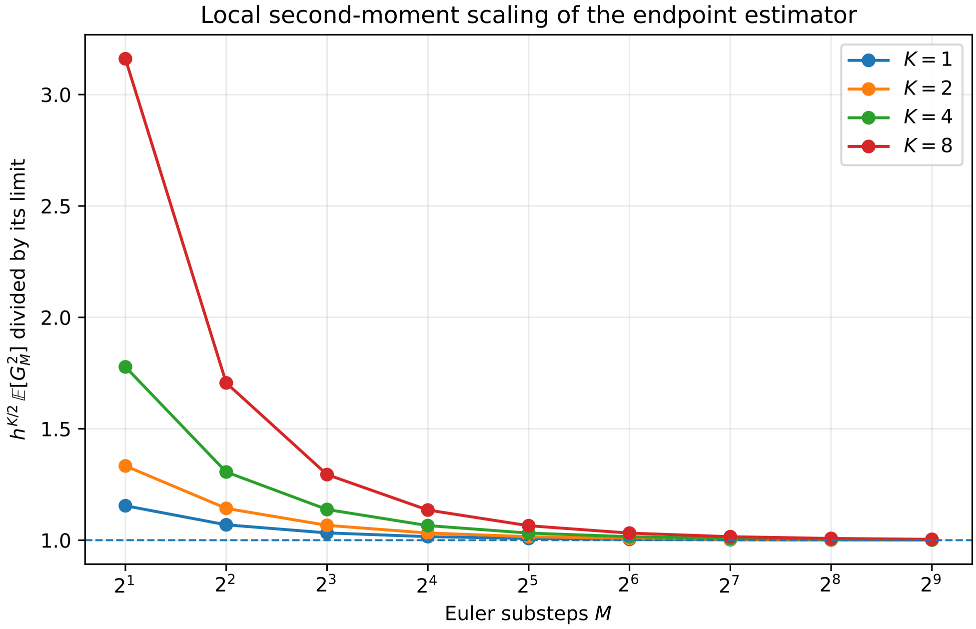

Figure 1: Exact local second-moment scaling for standard Brownian motion. The plotted quantity converges to one after division by its theoretical limit.

Table 4: Simulation of the four-dimensional Brownian counterexample path $M = S$. The estimator is divided by the true density.

M	Replications	Mean	Median	10th pct.	90th pct.	$\mathbb{P}(\widehat{q} < 0.5p)$
8	20000	1.001	0.3674	1.427×10^{-2}	2.940	0.560
16	20000	1.009	0.1582	1.520×10^{-3}	3.161	0.665
32	20000	1.027	0.03628	5.925×10^{-5}	2.746	0.754
64	20000	1.002	0.004223	8.507×10^{-7}	1.777	0.827
128	20000	1.042	1.537×10^{-4}	1.487×10^{-9}	0.7409	0.886
256	19531	0.941	1.118×10^{-6}	7.168×10^{-14}	0.1579	0.929
512	9765	0.998	8.604×10^{-10}	7.076×10^{-20}	0.01738	0.955

9.3 Typical estimates collapse while the mean remains correct

Figure 3 simulates the counterexample path $M = S$. The sample median falls rapidly towards zero, and by $M = 512$ more than 95% of estimates are below half the true density. The sample mean remains near one after normalisation because a small number of very large realisations carry the expectation.

9.4 A nonzero drift does not remove the dimensional law

Consider the diagonal Ornstein–Uhlenbeck process

$$dY_t = -Y_t dt + dW_t$$

in four dimensions. Its exact transition density is known, and the preterminal Euler state is Gaussian, so the second moment of the endpoint estimator can again be computed exactly. Figure 4 shows convergence to the same local constant in Theorem 4.2. The phenomenon is therefore not specific to driftless Brownian motion.

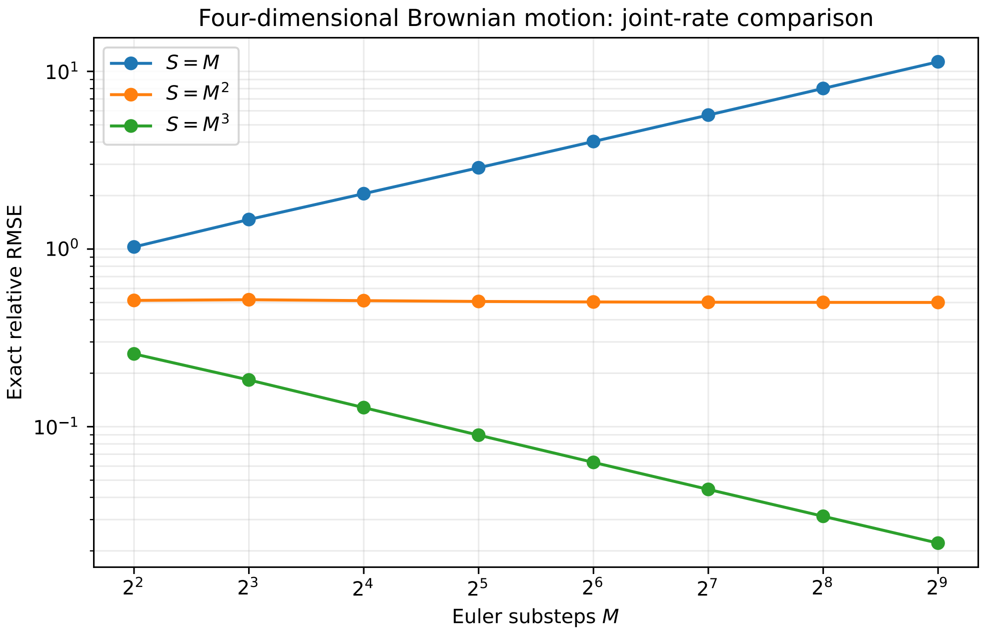

Figure 2: Exact relative RMSE at the origin for four-dimensional Brownian motion.

9.5 Euler boundary violations in the incompleteness process

Figure 5 evaluates (42) using the reported estimates $(\alpha, \beta) = (0.320, 0.088)$ and $(0.338, 0.101)$. Even with relatively fine steps, an ordinary Euler chain near the attainable zero boundary can become negative with non-negligible probability. Once it does, the next square-root coefficient is not real without an additional numerical convention.

As explained in Section 8.3, this calculation applies to the incompleteness specification (39) treated as a state in its own right. The implemented chain simulates (r, r^*, e, v) and recovers ϵ_t from (41), so the boundary that must not be crossed is the complete-markets surface rather than $\{X = 0\}$. The figure is therefore an exact illustration of the mechanism rather than of the implemented chain. The contrast with the interest rates in Table 2 is the substantive content: for r and r^* the Feller ratios exceed one by an order of magnitude and the boundary is never approached, whereas for the incompleteness state the ratio is exactly $1/2$ for all parameter values and, by Proposition 8.2, the boundary is reached exactly.

10 Remedies and alternative constructions

The endpoint variance problem is not an unavoidable property of likelihood inference for diffusions. It is a consequence of drawing preterminal states from the unconditioned forward Euler law and asking a shrinking final Gaussian kernel to connect them to a fixed observed endpoint.

10.1 Bridge and importance proposals

A bridge proposal guides simulated paths towards the observed endpoint. In importance-sampling language, it increases the probability of drawing paths in the region that contributes to the transition-density integral and corrects the proposal through a likelihood ratio. This is the direction taken by the acceleration methods studied by Durham and Gallant (2002),

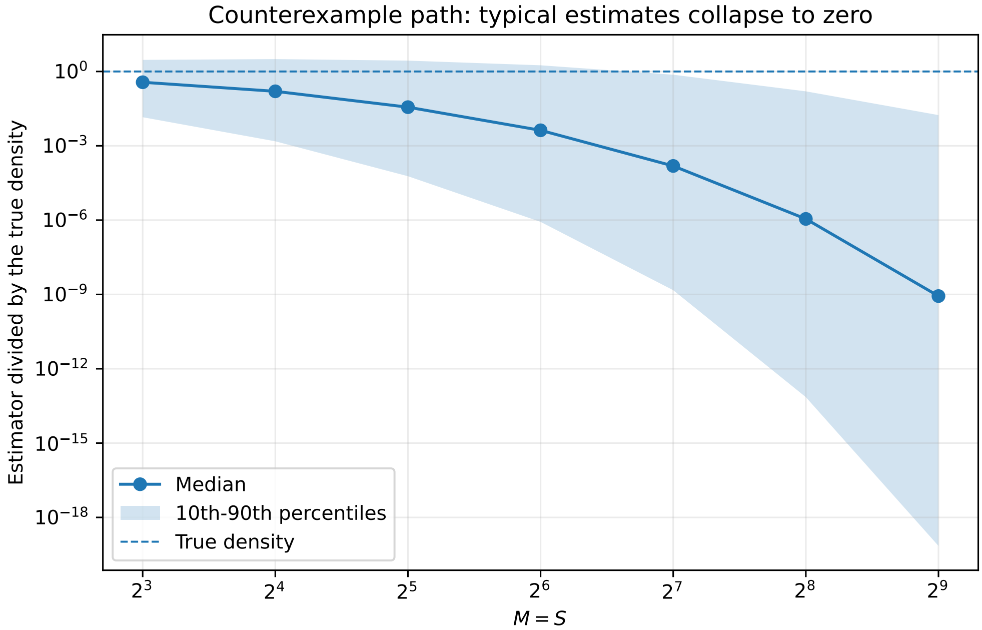

Figure 3: Monte Carlo distribution along $M = S$ for four-dimensional Brownian motion. Values are divided by the true density. The dashed line is one.

the path-augmentation schemes of Elerian et al. (2001), conditioned-diffusion methods such as Delyon and Hu (2006), the regularised proposal of Lindström (2012), and later guided proposals for multidimensional bridges (Schauer et al., 2017; Meulen and Schauer, 2017). The nonparametric simulation approach of Nicolau (2002) replaces the endpoint Gaussian kernel by a density estimate, which changes the bandwidth problem rather than removing it.

A useful proposal should be assessed through the moments of its importance weights. Merely changing the path simulator does not guarantee finite or dimension-stable variance, but conditioning on the endpoint directly targets the rare-event failure identified here. The numerical comparisons of Stramer and Yan (2007) are consistent with this reading: the endpoint estimator degrades as M grows at fixed S, which is the finite-sample signature of Corollary 4.3.

10.2 Exact or discretisation-free likelihood estimators

For restricted classes of diffusions, exact simulation and unbiased transition-density estimators remove Euler bias entirely. Beskos, Papaspiliopoulos, Roberts, and Fearnhead (2006) and Beskos, Papaspiliopoulos, and Roberts (2009) develop likelihood-based methods of this kind. Their applicability is model dependent, but they illustrate the correct theoretical structure: unbiasedness, continuity in parameters, finite-moment conditions, and joint control of data and Monte Carlo sample sizes are stated explicitly.

10.3 Analytical density expansions

Closed-form expansions, such as those of Aït-Sahalia (2002), avoid endpoint Monte Carlo variance but introduce an approximation-order problem of their own. They can be attractive for models that can be transformed to satisfy the required regularity conditions.

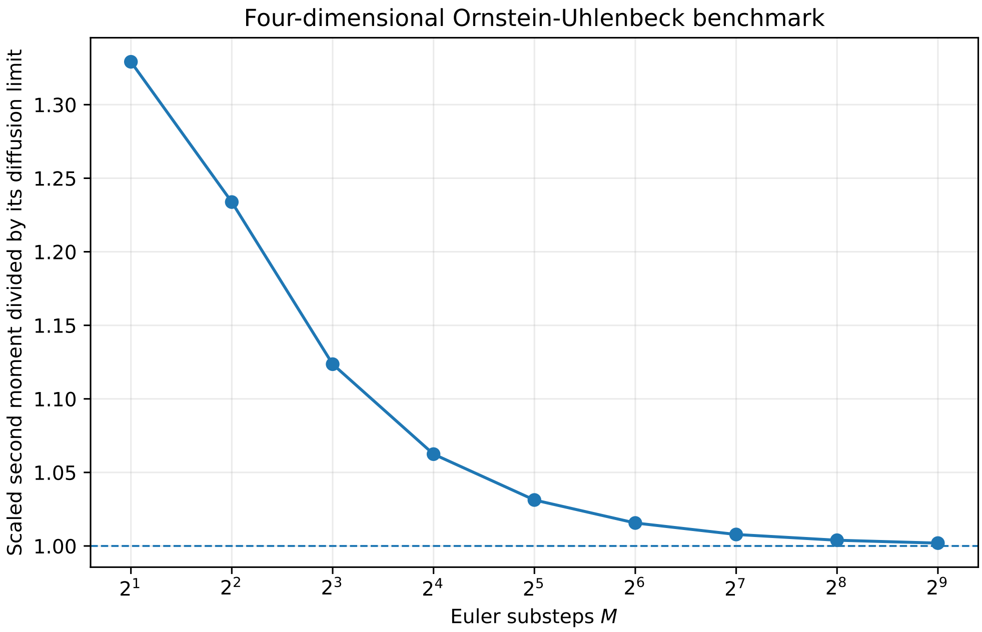

Figure 4: Scaled second moment for a four-dimensional Ornstein–Uhlenbeck diffusion.

10.4 Positivity-preserving discretisation

For square-root components, ordinary Euler should be replaced by an exact transition where available or by a specified positivity-preserving approximation. The choice is part of the estimator, not an implementation detail, because different truncation and reflection rules define different transition densities.

10.5 Practical design rule

If the unconditioned endpoint estimator is nevertheless used, M and S should be calibrated jointly. A minimum diagnostic is the effective endpoint size

$$R_{M,S} = S/M^{K/2}.$$

Increasing M without increasing $R_{M,S}$ is not an accuracy improvement. For density-level asymptotics, $R_{M,S}$ must diverge. For likelihood-level work, one should also monitor the variance and lower tail of every estimated transition density, use log-sum-exp evaluation, and verify stability after increasing both M and S along a theoretically admissible path.

11 Conclusion

The endpoint simulated-likelihood estimator has a simple but consequential dimensional structure. Its final Gaussian transition is a shrinking kernel. In K dimensions, the second moment of one simulated contribution grows as $M^{K/2}$, so the Monte Carlo variance is of order $M^{K/2}/S$. The relevant simulation size is $S/M^{K/2}$.

This leads to four definite conclusions about Brandt and Santa-Clara (2002). First, the proof of transition-density convergence combines two sequential limits without controlling their joint path. Second, the rate condition used in the asymptotic theorems is incompatible with mean-square density consistency in the four-dimensional application. Third, an exact Brownian

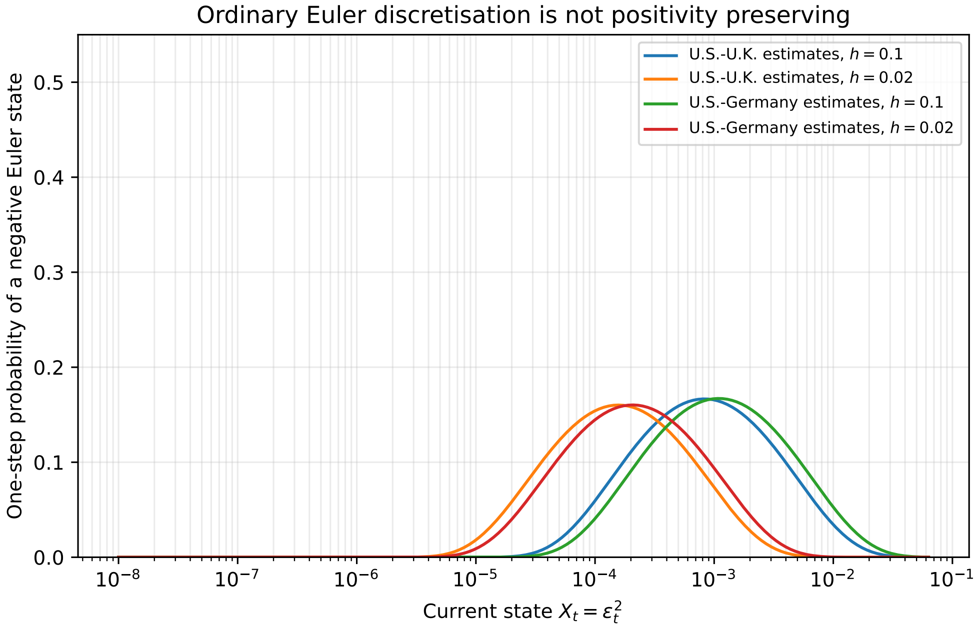

Figure 5: One-step probability that ordinary Euler discretisation of the incompleteness specification $X_t = \epsilon_t^2$ in (39) produces a negative state, using the estimates in Table 3 of Brandt and Santa-Clara (2002). The implemented estimator substitutes ϵ_t out and simulates (r, r^*, e, v) instead, so this is an illustration of the boundary mechanism rather than of the implemented chain; see Section 8.3.

sequence satisfying the stated condition, and satisfying the paper's own Assumptions 1 and 2, makes the simulated density converge in probability to zero rather than to the positive true density. The issue is therefore not only a missing technical lemma; the stated joint theorem is false. Fourth, and independently of all of the above, the implemented design $N = 544$, $M = 10$, $S = 5{,}000$ does not approach the region described by the paper's own conditions $\sqrt{S}/M \to 0$ and $N/S^{1/4} \to 0$, which would require $N^4 \ll S \ll M^2$.

The corrected analysis also supplies constructive results. The density estimator has a triangular-array CLT under $S/M^{K/2} \to \infty$. If the Euler density bias is first order, the MSE-optimal discretisation is $M \asymp S^{2/(K+4)}$. The nonlinear log-likelihood scale is $M^{K/2}/S$, subject to the usual lower-tail conditions required for expectations of logarithms. These results explain why bridge proposals and other importance-sampling methods are not merely computational refinements: they address the rare-event geometry of the estimator.

The exchange-rate application raises separate questions. Its square-root states fall outside the smooth uniformly elliptic theory, although for the interest rates the Feller ratios are large enough that this is a gap in proof coverage rather than a numerical hazard. The incompleteness state is different: its Feller ratio is exactly one half for every admissible parameter value, and the identification rule drives it onto its degenerate boundary exactly. Because the estimator substitutes that state out and simulates volatility instead, the requirement becomes that a recovered square root remain real along every simulated Euler path, which ordinary Euler discretisation does not guarantee. The volatility equation is implicit when risk prices depend on volatility, and the reported U.S.–U.K. estimates give a leading coefficient of the wrong sign, so that specification admits either no positive volatility branch or two, never exactly one. The Brownian correlation matrix requires a joint positive-semidefinite restriction that entrywise bounds do not deliver. The final empirical estimator also contains a minimum-incompleteness corner rule and binding constraints that are not covered by ordinary interior SML asymptotics.

None of these points, individually or collectively, proves that the finite-sample economic conclusions are false. They show that those conclusions cannot inherit the claimed maximum-likelihood asymptotics from the mathematical argument given. The appropriate post-publication response is a corrected theorem, a clear statement of the estimator actually implemented, and a re-examination of numerical stability under dimensionally valid simulation rates or bridge-based alternatives.

A Additional proof details

A.1 Tail control in the local moment theorem

For completeness, suppose that $V(z)$ is globally uniformly elliptic and bounded, μ is bounded, and $r_h(z \mid x) \le C$ uniformly. Let U be a neighbourhood of y and let $\delta = \inf_{z \notin U} \|z - y\| > 0$. For sufficiently small h, bounded drift gives

$$\|y - z - h\mu(z)\| \ge \delta/2 \qquad (z \notin U).$$

Uniform ellipticity then implies

$$g_h(y \mid z)^r \le C_r h^{-rK/2} \exp(-c_r/h) \qquad (z \notin U).$$

Multiplying by $h^{(r-1)K/2}$ and integrating against the probability density r_h gives a tail bound of order $h^{-K/2} e^{-c_r/h} \to 0$. This verifies the tail condition in Assumption 4.1 under one convenient global set of assumptions.

A.2 The Brownian collapse for more general subcritical paths

The proof of Theorem 3.3 extends beyond $M = S$. Let $h_n \downarrow 0$ and choose a radius

$$r_n = \sqrt{c h_n \log(1/h_n)}.$$

If

$$S_n h_n^{K/2} \{\log(1/h_n)\}^{K/2} \to 0$$

and S_n grows at most polynomially in $1/h_n$, then with high probability no preterminal draw lies within r_n of the endpoint, while the maximum kernel outside that ball tends to zero for sufficiently large c. Thus the estimator again collapses to zero. The theorem uses $M = S$ because it is transparent and directly satisfies the published rate in four dimensions.

B Reproducibility

The accompanying directory contains:

- `main.tex`, the manuscript source;
- `references.bib`, the bibliography;
- `code/generate_results.py`, the fully reproducible simulation and figure code;
- vector PDF and PNG versions of all figures;
- CSV and LaTeX versions of generated numerical tables.

The code uses a fixed seed, vectorised Gaussian simulation, and log-sum-exp evaluation of the endpoint estimator. No proprietary data are used.

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